

MITRE ATT&CK Game User Manual

MITRE ATT&CK Team:

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Installation & Setup

For Mac Users

1. Download and unzip the MacOS package.
2. Run the following command in the Terminal:

Shell

```
xattr -dr com.apple.quarantine MITREGame.app
```

3. Launch the game from the folder.

For Windows Users

1. Download and unzip the Windows OS package.
2. Run MITRE Game.exe to begin.

What is included in the downloaded file(s):

```
├─ MacOS/
│   ├── MITRE Game/          # Supporting assets for MacOS
│   ├── MITRE Game          # MacOS Application executable
│   └─ MITRE Game Teacher Resource Guide.pdf
├─ Windows OS/
│   ├── MITRE Game.exe       # Windows Application executable
│   └─ MITRE Game Teacher Resource Guide.pdf
├─ README.md                # Project overview and setup
└─ MITRE Game User Manual.pdf # Full system documentation
```

Interaction & Playing the Game

Welcome to the Kingdom of Cyborgia, Castellan! Your mission is to protect the realm from the Shadow Guild by deploying the right MITRE-based defenses using the King's Gold. The game progresses through three levels, with each getting progressively more difficult as the Guild launches more attacks.

Phase 1: The Build Phase

This is where the strategy happens. During this phase, the game is paused while you prepare your defenses.

- **Check the Intel Report:** Before you buy anything, click the **Intel Report** button in the top right. This report tells you what kind of tactics the Shadow Guild have headed your way so you can plan accordingly.
- **The Defense Cabinet:** On the left side of the screen, you'll see your available defenses. Click the (?) icon next to any defense to learn about its counter-technique.
- **Placing Defenses:** Click and drag a defense icon onto the castle map. If you drop it near its designated "socket," it will snap into place and deduct the cost from your **Available Funds**.
- **Removing Defenses:** Changed your mind? As long as you are still in the Build Phase, you can drag any placed defense in the trash can to remove it and get a full gold refund.

Phase 2: The Attack Phase

Once you feel ready, click the **LAUNCH ATTACK** button.

- **The Simulation:** You will see the Shadow Guild's attack(s) move toward your castle. The simulation lasts for 3 seconds while your automated systems attempt to block the incoming threats.
- **Persistence:** Remember, any defenses you build stay on the board for the next level! You can remove them from your castle, allowing you to create a massive, layered defense over time.

Phase 3: Mission Debrief

After the simulation ends, the server evaluates your performance.

- **The 70% Rule:** To pass a level, your combined defenses must cover at least 70% of the requirements for **every** attack launched in that round.
- **Winning:** If you pass, you'll receive a budget bonus and a health boost before moving to the next level.
- **Failing:** If your coverage is below 70%, the level fails. You will lose a chunk of health and must retry the level with a small "emergency" budget injection from the King to help you fix your strategy.

Gameplay Elements

1. Levels

- The game has three levels, each increasing in difficulty.
- The first level features one attack, the second features two, and the third features three.
- Your budget will increase with every level to account for the additional needed defenses.
- Upon failing a level, you will have to retry until you pass to advance to the next level.

2. Winning and Losing

- To pass a level, your combined defenses must cover at least 70% of the requirements for every attack launched in that round.
- Scoring less than 70% will result in losing the round and a loss of health.
- To win the game, you must successfully beat all three levels.
- A game loss occurs when a player's health points reach 0.

3. Defenses

- The defenses are listed in a scrollable cabinet on the left side of the screen.
- The defense's price is listed below its name and icon.
- Click on the question mark to open the defense's page in the Guidebook.
- To place a defense, click and hold down on the defense.
- A red square will appear on the castle. Drag the defense onto this square and let go.
- To remove a defense, drag it to the trash can. You will receive the funds back.
- When you are finished placing your defenses, click the LOCK DEFENSES button to start the attack phase.

4. Funds

- Player funds are shown at the top of the defense cabinet. Players start the game with \$1000.
- When a defense is placed down, the price is subtracted from the player's funds. A defense can only be placed if the player has sufficient funds.
- At the end of each level, players will receive more funds so they can afford more defenses.

5. Health and Shield

- The player starts with 100 health points, which are displayed as a health bar at the bottom of the screen.
- Failure to defend against attacks causes the player to lose health, while successfully defending against attacks causes the player to gain health.
- If a player gains additional health that exceeds 100 points, it is added to the shield bar.
- The shield bar functions as a second, bonus health bar.

6. Intel Report

- Click on the flashing Intel Report button to see a description of the coming attack, and use this information to decide which defenses you will place.
- The report updates every level as new attacks are added.

7. Reset Game

- Click the Reset Game button to go back to the Title Screen.

8. Guide

- Click the Guide button to open the Guidebook.

Things to Remember

1. **Health is Final:** Your health carries over between levels. If it hits zero, the kingdom falls and it's Game Over.
2. **Budget Wisely:** While Level 1 might feel easy, save some gold for the next levels as the Guild pulls out all the stops. Note that not all defenses are mapped to all of the attacks. Make sure to think critically about what you put down.
3. **Use the Guidebook:** If you're stuck, click the **GUIDE** button to see a full breakdown of every defense and how they function.
4. **Resting:** If you ever feel like you don't like the way your game is going, feel free to click the **RESET GAME** button in the lower right corner to get a fresh start.

Story Elements

Attacks Profiles: The Shadow Guild's Arsenal

1. The Poisoned Messenger (Phishing: T1566)

- **Narrative:** A courier delivers a letter that appears official, but the ink is infused with a toxin designed to incapacitate the very first individual who handles the parchment.
- **Technical Context:** This mimics **Phishing**, where attackers send fraudulent communications that appear to come from a reputable source. The goal is to steal sensitive data or install malware on the victim's machine.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1566/>

2. The Undercover Spy (Unsecured Credentials: T1552)

- **Narrative:** A member of the Shadow Guild infiltrates the castle in the guise of a servant, searching for physical keys left on hooks or eavesdropping on whispered secrets in the corridors.

- **Technical Context:** This represents the search for **Unsecured Credentials**. Attackers often look for passwords stored in plain text files, environment variables, or insecurely handled tokens within a network.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1552/>

3. The Corrupt Seneschal (Account Manipulation: T1098)

- **Narrative:** An enemy agent attempts to bribe the castle's record-keeper to alter the laws of the realm. Success allows the agent to grant themselves "Duke" status, providing them with unrestricted access to the castle gates.
- **Technical Context:** This mirrors **Account Manipulation**, specifically privilege escalation. Attackers modify account permissions to maintain access or gain higher-level administrative rights.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1098/>

4. The Trojan Crate (Obfuscated Files: T1027)

- **Narrative:** A merchant enters the castle with a crate of supposedly standard supplies. In reality, the crate houses a magical parasite engineered to spread through and weaken the castle's stone structure.
- **Technical Context:** This mimics **Obfuscated Files or Information**, where malicious code is hidden within seemingly benign files or through encryption to evade detection by security software.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1027/>

5. The Sorcerer's Script (Command and Scripting Interpreter: T1059)

- **Narrative:** An enemy mage stands beyond the castle walls, chanting ancient, complex commands. These incantations are designed to exploit the mechanical logic of the drawbridge, forcing it to lower against the will of the defenders.
- **Technical Context:** This represents the use of a **Command and Scripting Interpreter**. Attackers use scripts (like PowerShell or Python) to execute commands that bypass security controls and automate their attacks.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1059/>

6. The Bewitched Relic (User Execution: T1204)

- **Narrative:** The Guild leaves a beautifully crafted, seemingly harmless music box in the bustling courtyard. Overcome by curiosity, a young stablehand turns the crank, unwittingly triggering a dormant spell that unbars the barracks doors.
- **Technical Context:** Adversaries may rely on a user to execute a malicious file or run commands for initial access or lateral movement. This typically happens when a user

opens an email attachment, runs a macro in a document, or double-clicks a disguised executable file.

- **MITRE Link:** <https://attack.mitre.org/techniques/T1204/>

7. The False Raven (Adversary-in-the-Middle: T1557)

- **Narrative:** Shadow Guild rogues quietly intercept messenger birds flying between the kingdom's watchtowers. They meticulously swap the real scrolls with forged orders and send the birds on their way, manipulating the guard patrol routes from the shadows.
Technical Context: Adversaries may position themselves between two or more networked devices to intercept, monitor, alter, or redirect communications. This allows them to harvest sensitive credentials, session tokens, or manipulate payloads in transit.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1557/>

8. The Merchant's Disguise (Application Layer Protocol: T1071)

- **Narrative:** Shadow Guild spies masquerade as regular apple merchants and bards, passing secret hand signals and hidden messages right under the noses of the guards in the bustling town square.
Technical Context: Adversaries may communicate using application layer protocols to blend their command and control (C2) traffic with normal, legitimate network activity. Common protocols abused include HTTP, HTTPS, DNS, and SMTP.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1071/>

9. The Peasant Stampede (Endpoint Denial of Service: T1499)

- **Narrative:** The Guild pays a massive mob of fake peasants to crowd the drawbridge and riot at the gates, completely freezing castle operations and preventing legitimate royal messengers from entering or leaving.
- **Technical Context:** Adversaries may conduct Denial of Service (DoS) attacks targeted at endpoints to degrade, disrupt, or completely exhaust the availability of services, applications, or system resources, rendering them unavailable to legitimate users.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1499/>

10. The Smuggler's Tunnel (Exfiltration Over Alternative Protocol: T1048)

- **Narrative:** Instead of trying to carry stolen gold out through the heavily guarded main gate, thieves bypass the checkpoints by sneaking the kingdom's secrets out through sewer grates and underground rivers.
- **Technical Context:** Adversaries may steal or exfiltrate data from a victim network using a different protocol than the one originally used to compromise or control the system. This helps them bypass standard data loss prevention (DLP) filters.
- **MITRE Link:** <https://attack.mitre.org/techniques/T1048/>

Defense Profiles: The Royal Arsenal

1. The Two-Key Vault (Multi-factor Authentication: M1032)

- **Mechanism:** This defense requires both a physical wax seal and a secret whisper to gain entry. Even if a guard's physical keys are stolen by an intruder, the vault remains securely locked.
- **Cybersecurity Equivalent: Multi-factor Authentication (MFA).** It requires users to provide two or more verification factors to gain access to a resource.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1032/>

2. The Royal Guard (Privileged Account Management: M1026)

- **Mechanism:** These are elite soldiers assigned specifically to shadow high-ranking castle officials. Their sole purpose is to ensure that only the most trusted individuals in the kingdom handle the Royal Scepter.
- **Cybersecurity Equivalent: Privileged Account Management (PAM).** This involves monitoring and protecting accounts that have elevated permissions to prevent unauthorized use.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1026/>

3. The Portcullis Toll (Web Restrict: M1021)

- **Mechanism:** A checkpoint established at the very edge of the kingdom's borders. It is designed to stop merchants arriving from known "enemy lands" and to forbid the entry of strange or unidentified crates.
- **Cybersecurity Equivalent: Web Restrict.** This includes web filtering and firewalls that block access to malicious domains or prevent the download of unauthorized content.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1021/>

4. Iron-Bound Chests (Permissions: M1022)

- **Mechanism:** Sensitive scrolls and documents are locked within specific rooms of the keep. Under this system, a cook has no need for a key to the armory, and a stablehand is denied access to the Royal Treasury.
- **Cybersecurity Equivalent: File and Directory Permissions.** This follows the principle of "least privilege," ensuring users only have access to the data necessary for their specific roles.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1022/>

5. The Kennel Hounds (Antivirus/Antimalware: M1049)

- **Mechanism:** These beasts are trained to detect the scent of poisoned items, hidden daggers, or any "unnatural" objects brought into the citadel.

- **Cybersecurity Equivalent: Antivirus (AV).** AV software scans for known signatures of malicious software and monitors for suspicious behavior on a system.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1049/>

6. Stone Wall Reinforcement (OS Hardening: M1028)

- **Mechanism:** Specialized laborers fill in ancient cracks in the masonry, seal off unused drainage pipes, and remove hidden "thief holes" that have existed for generations.
- **Cybersecurity Equivalent: Operating System (OS) Hardening.** This process involves securing a system by reducing its surface of vulnerability, such as by disabling unused services or closing unnecessary ports.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1028/>

7. The Scribe's Ledger (Audit: M1047)

- **Mechanism:** A dedicated monk stationed at the gates meticulously records every individual who enters, leaves, or moves between the different rooms of the castle. While this ledger cannot physically stop a crime, it provides a perfect record of exactly who was involved after the event occurs.
- **Cybersecurity Equivalent: Audit/Logging.** Detailed logs allow security teams to track the "who, what, and when" of a system breach for forensic analysis.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1047/>

8. Village Drills (User Training: M1017)

- **Mechanism:** Regular town hall meetings are held where the peasantry is educated on common ruses. They are taught that the King will never use a random messenger bird to request their gold coins.
- **Cybersecurity Equivalent: User Training.** Security awareness training helps employees recognize phishing attempts and social engineering tactics.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1017/>

9. The Scribe's Vault (Data Backup: M1053)

- **Mechanism:** Keeps reserve copies of the kingdom's records so losses can be restored after an attack.
- **Cybersecurity Equivalent: Data Backup.** Creates restorable copies of critical data.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1053/>

10. The Outland Post (Remote Data Storage: M1029)

- **Mechanism:** Stores copies far from the castle so a local disaster cannot destroy everything at once.

- **Cybersecurity Equivalent: Remote Data Storage.** Off-site data storage to prevent local loss.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1029/>

11. The Royal Inquisitor (SSL/TLS Inspection: M1020)

- **Mechanism:** Examines sealed deliveries for hidden dangers before allowing them into the keep.
- **Cybersecurity Equivalent: SSL/TLS Inspection.** Decrypts traffic to find hidden threats.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1020/>

12. The Royal Lineage (Active Directory Configuration: M1015)

- **Mechanism:** A strict hierarchy of nobles and titles is enforced. A mere squire cannot simply declare themselves a Duke to command the castle guards or access the treasury.
- **Cybersecurity Equivalent: Active Directory Configuration.** Configures directory services securely to prevent privilege escalation and credential theft.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1015/>

13. The Steward's Roster (User Account Management: M1018)

- **Mechanism:** The Steward actively revokes castle access from mercenaries whose contracts have expired and banishes idle loiterers from the courtyard.
- **Cybersecurity Equivalent: User Account Management.** Manages the lifecycle of user accounts to limit unnecessary or stale privileges.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1018/>

14. The Riddler's Gate (Password Policies: M1027)

- **Mechanism:** Guards demand a complex, ever-changing passphrase from travelers—no more using "password" or the King's birthday to gain entry.
- **Cybersecurity Equivalent: Password Policies.** Enforces strong password complexity, length, and rotation requirements.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1027/>

15. The Walled Districts (Network Segmentation: M1030)

- **Mechanism:** The castle is divided into isolated rings. Even if the outer bailey falls to the Shadow Guild, the inner keep remains sealed behind raised drawbridges.
- **Cybersecurity Equivalent: Network Segmentation.** Divides a network into smaller, isolated segments to contain lateral movement during a breach.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1030/>

16. The Tower Ballistas (Network Intrusion Prevention: M1031)

- **Mechanism:** Vigilant archers actively scan the horizon, immediately shooting down any incoming projectiles, siege weapons, or suspicious ravens before they land.
- **Cybersecurity Equivalent: Network Intrusion Prevention.** Monitors network traffic to actively detect and block malicious exploits.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1031/>

17. The Blacksmith's Seal (Limit Software Installation: M1033)

- **Mechanism:** Only tools and weapons forged by the official royal armory are allowed. Peasants are forbidden from bringing their own makeshift swords into the barracks.
- **Cybersecurity Equivalent: Limit Software Installation.** Restricts users from installing unauthorized software applications.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1033/>

18. The Privy Council (Limit Access to Resource Over Network: M1035)

- **Mechanism:** Only designated, high-ranking advisors are allowed into the map room; the doors are completely hidden and walled off from the rest of the kingdom.
- **Cybersecurity Equivalent: Limit Access to Resource Over Network.** Restricts access to specific network resources based on logical business needs.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1035/>

19. The Customs Officers (Filter Network Traffic: M1037)

- **Mechanism:** Guards rigorously inspect all incoming merchant carts and outgoing messengers, turning away anyone lacking the proper royal travel permits.
- **Cybersecurity Equivalent: Filter Network Traffic.** Filters incoming and outgoing network traffic based on IP addresses, ports, or protocols.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1037/>

20. The King's Decree (Execution Prevention: M1038)

- **Mechanism:** A magical ward that instantly paralyzes anyone attempting to cast unauthorized spells or read from forbidden tomes within the castle walls.
- **Cybersecurity Equivalent: Execution Prevention.** Blocks unauthorized scripts, binaries, or untrusted code from executing on endpoints.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1038/>

21. The Suspicion Ward (Behavior Prevention on Endpoint: M1040)

- **Mechanism:** Guards closely monitor the behavior of the staff. If the court jester suddenly starts drawing a sword instead of juggling, he is tackled immediately.
- **Cybersecurity Equivalent: Behavior Prevention on Endpoint.** Monitors and blocks active processes exhibiting suspicious or anomalous behavior patterns.

- **MITRE Link:** <https://attack.mitre.org/mitigations/M1040/>

22. The Ciphered Scrolls (Encrypt Sensitive Information: M1041)

- **Mechanism:** Royal decrees are written in a dead, magical language. Even if a Shadow Guild spy steals the scroll, they cannot read a single word of it.
Cybersecurity Equivalent: Encrypt Sensitive Information. Encrypts sensitive data at rest or in transit to protect confidentiality from unauthorized eyes.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1041/>

23. The Sealed Catacombs (Disable or Remove Feature or Program: M1042)

- **Mechanism:** Laborers purposely collapse old, forgotten tunnels and board up abandoned secret passages so thieves cannot use them to bypass the gates.
- **Cybersecurity Equivalent: Disable or Remove Feature or Program.** Disables or removes unnecessary software features or protocols to reduce the overall attack surface.
MITRE Link: <https://attack.mitre.org/mitigations/M1042/>

24. The Royal Wax Seal (Code Signing: M1045)

- **Mechanism:** Every official missive must bear the King's enchanted wax seal. Forgeries lacking the proper crest are immediately thrown into the hearth.
- **Cybersecurity Equivalent: Code Signing.** Digitally signs software to verify the identity of the author and ensure it hasn't been altered.
MITRE Link: <https://attack.mitre.org/mitigations/M1045/>

25. The Mason's Upkeep (Update Software: M1051)

- **Mechanism:** Stonemasons constantly patch weathering walls, reinforce rusting iron gates, and upgrade the defenses against newly discovered Shadow Guild siege tactics.
- **Cybersecurity Equivalent: Update Software.** Applies security patches to software and operating systems to remediate known vulnerabilities.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1051/>

26. The Keep's Protocols (Software Configuration: M1054)

- **Mechanism:** Strict rules govern the castle's daily operations. Fires must be extinguished at night, and weapons must be locked away, ensuring the castle operates safely.
- **Cybersecurity Equivalent: Software Configuration.** Configures software applications securely to minimize corporate risk and enforce safety baselines.
- **MITRE Link:** <https://attack.mitre.org/mitigations/M1054/>

Different Game Modes

The game features three distinct gameplay modes tailored to different learning objectives, skill levels, and strategic playstyles: **Beginner, Random, and Custom.**

- **Beginner Mode:** Features a structured, three-level simulation where players defend against a tiered progression of attacks that gradually increase in difficulty and threat complexity. Using a set of 11 defenses and 5 attacks, players manage an initial budget and must strategically purchase and commit to defenses (including decoys) before viewing attack outcomes.
- **Random Mode:** Maximizes replayability by introducing unpredictability into every session. The mode starts players with 10 initial defenses, then randomly shuffles and unlocks 5 defenses while generating 1 to 3 random attacks per level. To ensure fairness, the system dynamically calculates budgets and validates attack combinations so every level remains winnable, automatically resetting the environment and generating a new scenario upon completion.
- **Custom Mode:** Designed for educators and students looking to target specific cyber threats and mitigations. This mode provides full control by allowing players to manually select the exact attacks and defense systems in play. The system verifies strategy viability and dynamically computes budgets to guarantee that custom scenarios are balanced and beatable.

Edge Cases

Sometimes, even the most legendary Castellans try to break the rules of engagement. Don't worry—the Game Server is always watching and will prevent any illegal moves that don't follow the manual's protocols.

Here are a few common edge cases where the system will step in and keep you on the right path:

- If you try to drag a high-tier defense onto the board when you're low on gold, the server will block the placement. You can only snap a defense into its socket if you have enough **Available Funds** to cover the cost.
- Strategy is for the **Build Phase**, not the **Attack Phase**. If you try to place or move a defense while the Shadow Guild's attacks are already in flight, the server will deny the action. You must wait for the simulation to end and return to the Build Phase to make changes.
- If you drop a defense icon in the middle of a forest instead of a designated castle socket, it won't just sit there. The icon will snap back to your cabinet immediately so you don't accidentally waste gold on a defense that isn't actually protecting anything.

- Trying to refund a defense during the Attack Phase to save your gold? The server only allows for removals and refunds during the **Build Phase**; once the Shadow Guild is at the gates, your defenses are locked in.

Castellan's Tip: When the server blocks an action, it's usually because you're trying to do something outside of the current game state. Check your status in the sidebar to see if you're in the Build or Attack Phase!

Scalability & Inheritance

The Kingdom of Cyborgia is built to grow! Our system's architecture is designed for easy scalability, allowing future developers to extend the game beyond the initial three levels to create an even longer campaign. Because the game uses a flexible backend, adding new MITRE ATT&CK techniques or additional defenses is straightforward, as they can inherit the logic already established in the current framework. We also envision the game evolving visually. Getting professional graphic design help in the future would allow for custom sprites and more detailed environments, making the simulation even more immersive for every Castellan who takes the throne.

Additional Materials

- Teacher Guide
- README